

Next Generation Data Centers and Energy Management Systems

Lecture 5B: Liquid Cooling & Advanced Heat Transfer

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Learning Plan

- Introduction to Data Center Systems and Thermodynamics
- Psychrometrics and Heat Transfer
- Performance Metrics and Standards
- Applied Computational Fluid Dynamics (CFD)
- **Liquid Cooling & Advanced Heat Transfer**
- Chip Cooling: Heat Conduction and Thermal Interface Materials
- Power Delivery and Infrastructure (Energy sources: Geothermal, Solar, Nuclear, etc.)
- Overview of Simulation Field; the Concept of a Digital Twin
- Machine Learning and Optimization Tools
- Modeling and Simulation of an Energy Management System
- Control Systems and Optimization
- Spacecraft Thermal Control
- Final Project Work, Peer Feedback, and Mentor Check-in
- Symposium

Learning Objectives

- Explain the physical limitations of air cooling in high-TDP (Thermal Design Power) environments.
- Identify the key components used to transfer heat away from a chip (TIM, Spreaders, Heat Pipes).
- Differentiate between various liquid cooling architectures (Cold Plates vs. Immersion).
- Describe the role and function of a Cooling Distribution Unit (CDU) in a data center environment.

The Heat Challenge: Why Liquid?

- **Power Density:** Increasing TDP (Thermal Design Power) leads to extreme heat fluxes at the chip level.
- **Limitations of Air:**
 - Low volumetric heat capacity.
 - Requires large volumes of air and high fan speeds (power intensive).
 - Limited heat transfer coefficient (α).
- **The Liquid Advantage:**
 - Much higher thermal conductivity and heat capacity.
 - Allows for much higher heat flux removal in a smaller footprint.

Thermal Interface Materials (TIM)

- **The Problem:** Surface roughness at the chip/spreader interface creates microscopic air gaps (insulators).
- **The Solution:** TIMs (greases, pads, liquid metal) fill these gaps to create a continuous conduction path.
- **Key Metrics:**
 - **Thermal Conductivity (k):** Higher is better.
 - **Bond Line Thickness (BLT):** Minimizing thickness reduces total thermal resistance (R_{th}).

Thermal Resistance Equation

$$R_{TIM} = \frac{t}{k \cdot A}$$

Spreading the Heat: Spreaders & Heat Pipes

- **Heat Spreaders:**

- High-conductivity metal (usually Copper) to distribute heat laterally.
- Reduces “spreading resistance.”

- **Heat Pipes & Vapor Chambers:**

- Use **phase change** (evaporation/condensation) to transport heat.
- **Heat Pipe:** 1D transport via capillary action.
- **Vapor Chamber:** 2D spreading via phase change; extremely efficient for high-flux chips.

Liquid Cooling Architectures: Cold Plates

- **Direct-to-Chip (Cold Plate) Cooling:**

- A liquid-cooled metal block is mounted directly to the chip.
- Liquid flows through internal channels to absorb heat.

- **Microchannel Technology:**

- Etching tiny channels into the cold plate to vastly increase the surface area.
- Maximizes the convective heat transfer coefficient (h).

Liquid Cooling Architectures: Immersion

- **The Concept:** Submerging the entire IT hardware in a dielectric (non-conductive) fluid.
- **Single-Phase Immersion:**
 - Fluid remains liquid; cooled via convection and circulated by pumps.
- **Two-Phase Immersion:**
 - Fluid boils at the chip surface.
 - Utilizes **latent heat of vaporization** for superior cooling.

Case Study: Aquasar Technology

- **Concept:** A hot water cooled supercomputer prototype designed for direct energy reuse.
- **Performance Comparison:**
 - **Liquid (Hot Water):** $\Delta T_{chip/coolant} \approx 15^{\circ}\text{C}$ (at 60°C coolant).
 - **Air Cooling:** $\Delta T_{chip/air} \approx 35^{\circ}\text{C}$ (requiring pre-cooling to 23°C).
- **Key Advantages:**
 - Eliminates the need for energy-intensive chillers.
 - Enables **Heat Reuse**: 60°C water can be used for building heating.
 - **Efficiency**: Achieved 80% heat recovery efficiency and 34% exergetic efficiency.
- **Economic Insight:** Space heating provides the highest economic value for recovered heat.

System-Level: Cooling Distribution Unit (CDU)

- **The Role:** Manages the heat exchange between the facility water and the IT cooling loop.
- **Two-Loop System:**
 - **Primary Loop:** Facility water (chilled water from the building).
 - **Secondary Loop (TCS):** Treated/dielectric fluid that circulates through the servers.
- **Core Functions:**
 - Heat Exchange (Plate Heat Exchangers).
 - Pumping and Flow Control.
 - Filtration and Water Chemistry Management.
 - Leak Detection and Safety Monitoring.

Radiative Cooling in Space Systems

- **The Challenge:** In the vacuum of space, there is no atmosphere for convection or conduction to a medium. **Radiation** is the only mechanism for heat rejection.
- **The Thermal Path:**
 - High-power chips → Liquid/Pumped Loops → External Radiator Panels → Deep Space.
- **Physics (Stefan-Boltzmann Law):**

$$Q_{rad} = \epsilon \sigma A (T_{rad}^4 - T_{space}^4)$$

- **Key Design Considerations:**
 - **High Emissivity (ϵ):** Using specialized coatings to maximize heat rejection.
 - **Radiator Area (A):** Large surface areas are required due to the T^4 dependence.
 - **Solar/Albedo Loading:** Managing heat from the Sun or Earth that hits the radiator.
 - **Mass Constraints:** Every kg of radiator/fluid adds to launch cost.

Typical layout of a data center facility

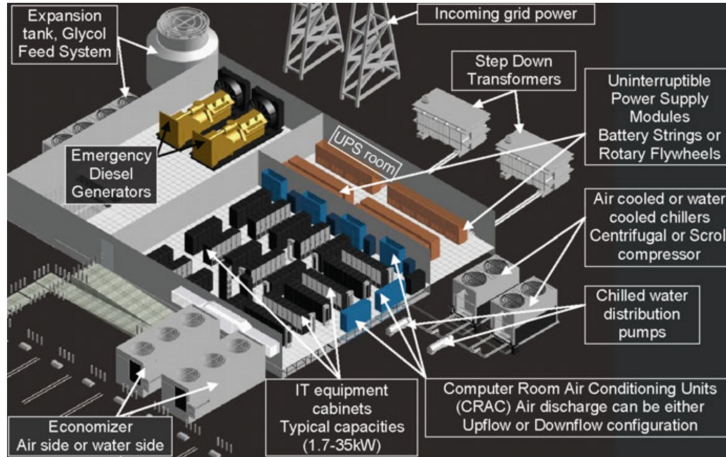


Figure 1: (Figure 1.1 in Ref. [1]).

Seminar 2

- Dr. David Geb, Synopsys (formerly Ansys)
- Title: TBD
- Date: Tuesday, 9/29/26
- Time: 15:30 (Berkeley Time)
- Location: GSPP 150

Microelectronics cooling: Solution to Blasius

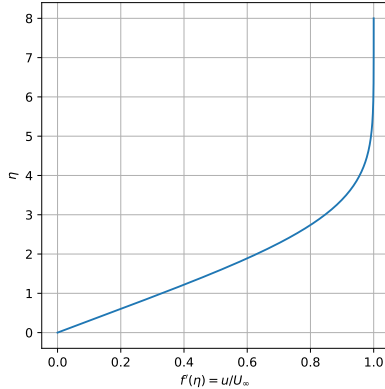


Figure 2: Blasius profile. Solved numerically in 1908. Read his biography [here](#).

Microelectronics cooling: Solution to Pohlhausen

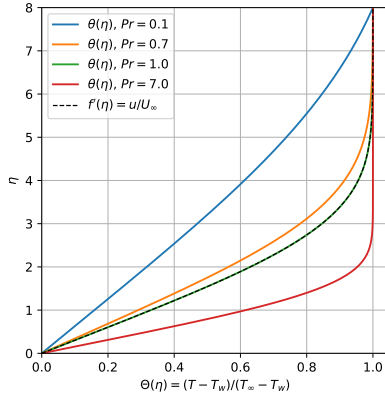


Figure 3: Solution to the Pohlhausen equation. Read his biography [here](#).

References I

- [1] Yogendra Joshi and Pramod Kumar, editors.
Energy Efficient Thermal Management of Data Centers.
Springer US, 2012.